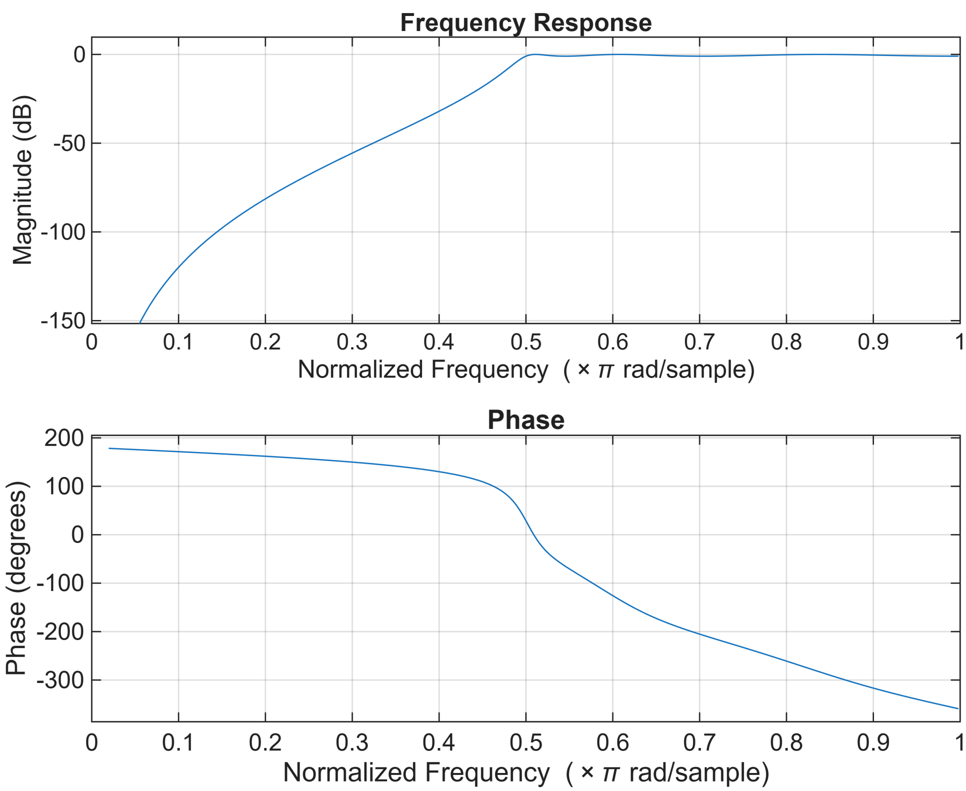
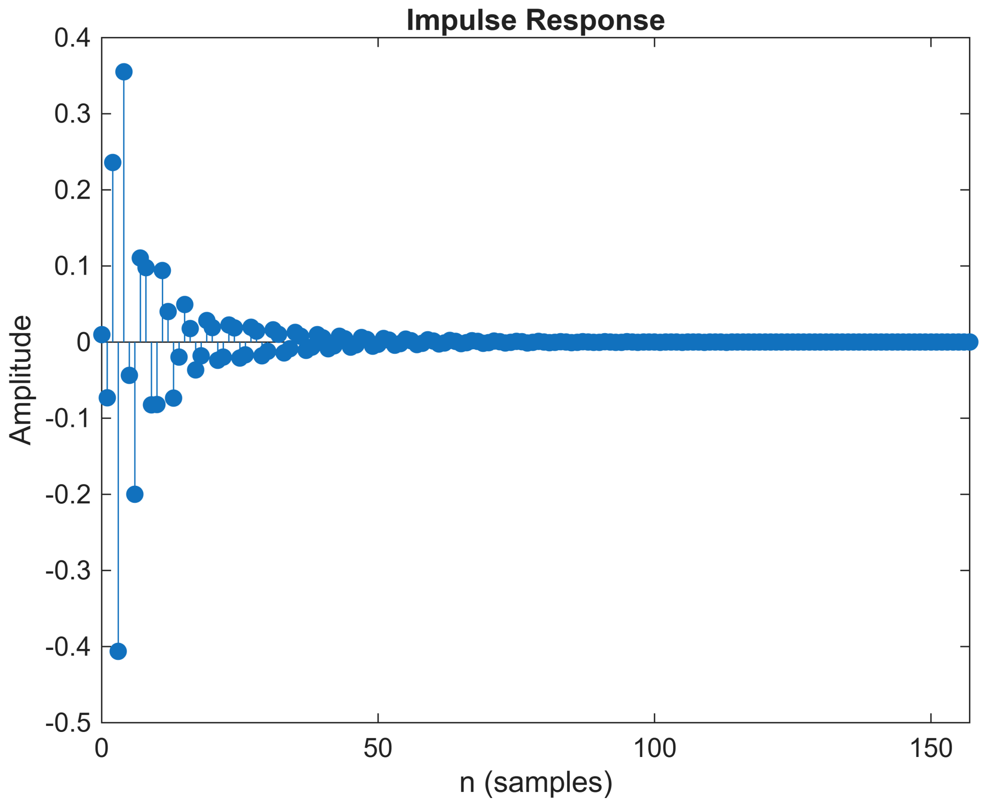
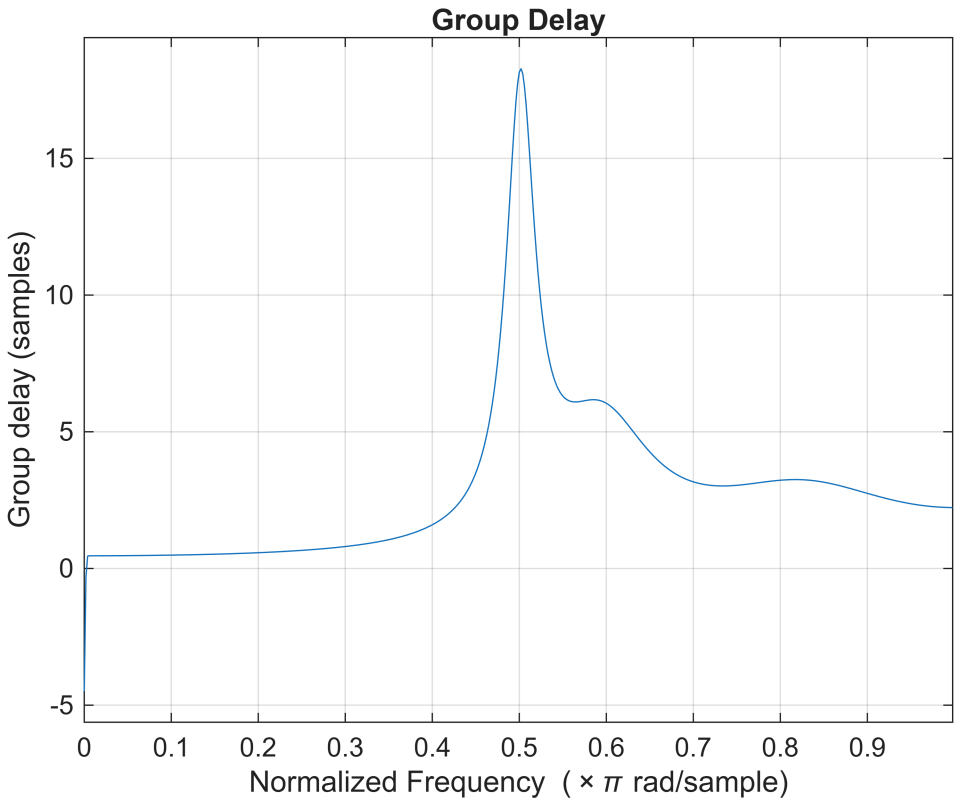
```
clc;
clear;
close all;

% Specifications
Rp = 1;       % Passband Ripple (dB)
Rs = 40;      % Stopband Attenuation (dB)
Wp = 0.5;     % Passband Edge
Ws = 0.35;    % Stopband Edge

% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');
```



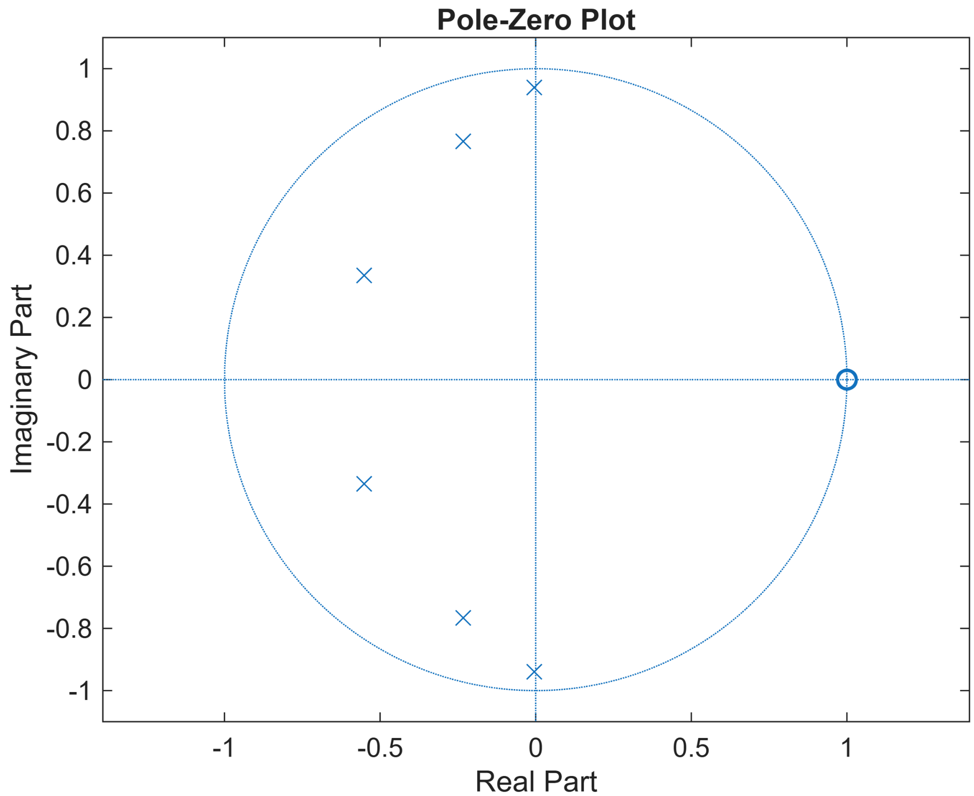

```

% Impulse Response
figure;
impz(b,a);
title('Impulse Response');
```




```
% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');
```




```
% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');
```




```
fprintf('Filter Order = %d\n',n);
```

```
Filter Order = 6
```

```
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

```
Cutoff Frequency = 0.5000
```